

Pranav Balaji Kapratwar

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Summary

Final-year B.Tech Computer Science (Data Science) student with expertise in Full Stack Development, Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing (NLP), and Generative AI. Experienced in building end-to-end AI applications using Python, NumPy, Pandas, Flask, FastAPI, LangChain, FAISS, Grok API, TensorFlow, Streamlit, and SQL. Skilled in Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), prompt engineering, REST API development, model deployment, and scalable backend systems, with a strong foundation in Data Structures, OOP, DBMS, and problem-solving. Quick learner with a strong interest in expanding into Computer Vision and Generative AI architectures.

Skills

Languages	Python, SQL
Backend	Flask, FastAPI, REST API
Frontend	HTML, CSS, Streamlit
ML / AI	Machine Learning, Deep Learning, NLP, Scikit-Learn, TensorFlow, NLTK, Pandas, NumPy, Prompt Engineering, Large Language Models (LLMs), RAG, LangChain, Generative AI
Familiar With	PyTorch, ANN, CNN, RNN, Computer Vision, YOLO, Hugging Face Transformers
Database	MySQL, SQLite, DBMS
Cloud	AWS
Mathematics	Statistics, Probability, Linear Algebra
Core CS	OOP, DSA, OS Basics, DBMS
ML Practices	Feature Engineering, Hyperparameter Tuning, Model Deployment, Experimentation, Cross-Validation
Tools	GitHub, Git, VS Code, BentoML, Pickle, Docker
Soft Skills	Communication, Leadership, Time Management, Presentation

Projects

StudyBot — AI-Powered RAG Study Assistant 2026

Python · Streamlit · LangChain · Grok API · FAISS · SQLite · NLP · RAG

github.com/Pranavkapratwar10/studybot

- Developed a secure AI-powered study assistant with user authentication, allowing students to upload multiple study materials and ask questions in natural language.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, and Grok API to retrieve relevant document content and generate context-aware answers.
- Implemented five exam preparation modes including Viva, MCQs, Short Answers, Long Answers, and Revision Notes, along with intelligent fallback search when relevant content was unavailable.
- Designed an admin dashboard with SQLite-based user management, persistent chat history, document tracking, and system statistics for monitoring application usage.

TalentTrack — Campus & Corporate Hiring Platform 2025

Flask · HTML · CSS · SQLite · pdfplumber · Tesseract OCR · REST API · NLP

github.com/Pranavkapratwar10/Talenttrack1

- Built a dual-panel web app using Flask and SQLite allowing admins to post jobs and students to register, upload resumes, and apply from one portal.
- Developed a resume parser using pdfplumber and Tesseract OCR to extract structured data from any PDF format including scanned files.
- Designed an ATS ranking engine using NLP similarity scoring to rank applicants automatically.
- Implemented email notification system for selection/rejection with reasons and automated vacancy cap enforcement via REST API logic.

Traconomy — Personal Finance Management System 2025

Python · Flask · HTML · CSS · SQLite · REST API

github.com/Pranavkapratwar10/traconomy

- Built a full-stack finance tracker with expense logging, savings goals, and custom categories using Flask and SQLite.
- Implemented downloadable Excel and PDF report generation for financial summaries.

- Integrated automated email alerts for goal completion and priority-based financial event reminders.
- Added EMI, SIP, FD, and Tax calculators alongside calendar-based monthly performance tracking.

AI-Powered Student Feedback Analyzer

Sept – Oct 2025

Python · Scikit-Learn · NLP · SVM · Random Forest · Streamlit
github.com/Pranavkapratwar10/my-project-student-feedback-analyzer

- Developed an NLP pipeline processing multilingual feedback, achieving 92% classification accuracy while reducing manual review time.
- Designed a sensitive-issue detection module to flag harassment and discrimination in feedback data.
- Created an interactive Streamlit dashboard for trend visualization and automated report generation.
- Applied tuned SVM and Random Forest models to maximize classification performance.
- Secured Runner-Up at Technovation 1.0 Hackathon among 100+ teams.

Internship

SaiKet Systems

Oct 2025 – Nov 2025

Machine Learning Intern

- Processed and cleaned an unstructured hotel management dataset using Python and Pandas, handling missing values, inconsistent formats, and noise.
- Performed feature engineering and exploratory data analysis (EDA) to uncover booking patterns, cancellation trends, and seasonal demand insights.
- Trained a Random Forest classifier for cancellation prediction; improved performance via hyperparameter tuning and cross-validation.
- Achieved high predictive performance after hyperparameter tuning and cross-validation, demonstrating strong model generalization.

Education

B.Tech Computer Science (Data Science)

2023 – Present

Kolhapur Institute of Technology

CGPA: 8.2

Class XII — Maharashtra Board

2023

Shahu Maharaj College, Loha

Percentage: 71.5%

Class X — Maharashtra Board

2021

Nalge High School, Loha

Percentage: 92.2%

Achievements

- Runner-Up — Technovation 1.0 Hackathon among 100+ competing teams.
- Published dataset on Kaggle with 14K+ views and 3K+ downloads.
- Solved 130+ problems on LeetCode focusing on core DSA patterns.

Certifications

- 100 Days of Code: Complete Python Pro Bootcamp Oct 2024 – Jan 2025
- Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp Jan 2026 – Mar 2026

Languages

English — Hindi — Marathi